to measure the value of physical properties that can vary continuously over a range. Values of discrete data types only have meaning as individual values and SHALL NOT be added or subtracted.

Some data types specify bit-widths for future potential growth in range (analog) or number of values (discrete).

Cluster specifications SHALL use the data type short name to reduce the text size of the specification.

Data types can be defined in three locations:

1. Cluster specification.
2. Cluster category sections.
3. Data Model.

The data types defined in the Data Model are the shareable data types and MAY be used within any cluster specification. The data types defined in the cluster category sections are the shareable data types within that specific cluster category and SHALL only be used by clusters within that cluster category. The data types defined in a cluster specification only apply to that specific cluster and SHALL NOT be reused in other cluster specifications.

The data types defined in the Data Model and cluster category section SHALL NOT have naming conflicts. A data type defined in the cluster specification MAY have the same name as a data type defined in the Data Model or cluster category section.

To determine which data type is used by an element, the following rules apply, in order from top to bottom:

- If the data type is defined in the cluster specification, this is used.
- If the data type is defined in the cluster category section and not in the cluster specification within that cluster category section, this is used.
- If the data type is not defined in the cluster specification or the cluster category section, the data type defined in the Data Model is used.

7.19.1. Base Data Types

Class	Data Type	Short	Size
Discrete	**Boolean**		
	Boolean	**bool**	1 byte
	Bitmap		
	8-bit bitmap	**map8**	1 byte
	16-bit bitmap	**map16**	2 bytes
	32-bit bitmap	**map32**	4 bytes
	64-bit bitmap	**map64**	8 bytes

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